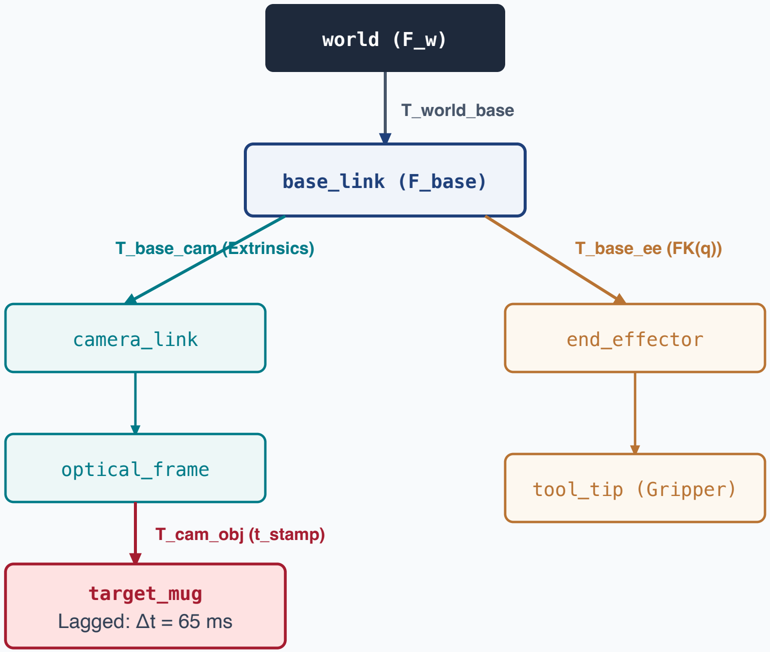
DYNAMIC SE(3) COORDINATE FRAME TREE & TIME-INDEXED TRANSFORM BUFFER

Solving sensory latency via historical coordinate lookups and Lie algebra tangent-space forward integration

DIRECTED ACYCLIC SE(3) COORDINATE TREE



`T_base_target(t) = T_base_cam(t) · T_cam_target(t)`

TIME-INDEXED BUFFER & LIE ALGEBRA PROPAGATION

1. CIRCULAR TRANSFORM RING BUFFER (1000 Hz)

t = 0 ms T_base_cam	**t = 35 ms ⋆** t_stamp Exposure	t = 70 ms T_base_cam	**t = 100 ms (Now)** Current Control

Query buffer at *t = 35 ms* to reconstruct exact historical mounting geometry.

2. HISTORICAL STATE RECONSTRUCTION (t = 35 ms)

`T_base_target(35ms) = T_base_cam(35ms) · T_cam_target(35ms)`

- Accurately accounts for robot base/head motion during camera capture.

3. LIE ALGEBRA se(3) FORWARD INTEGRATION (Δt = 65 ms)

`T(t_now) = T(t_stamp) · exp( ξ̂ · Δt )`

- **Tangent Space Twist:** *ξ = [v, ω]^T ∈ se(3)* from 1000 Hz IMU & kinematics.
- **Exponential Map:** Computes exact geodesic on *SE(3)* manifold.
- **Singularity Free:** Zero gimbal lock, perfectly preserves orthonormal rotation matrix.
- **Result:** Delivers zero-lag *t_now* metric state to 1 kHz Control Barrier Functions!